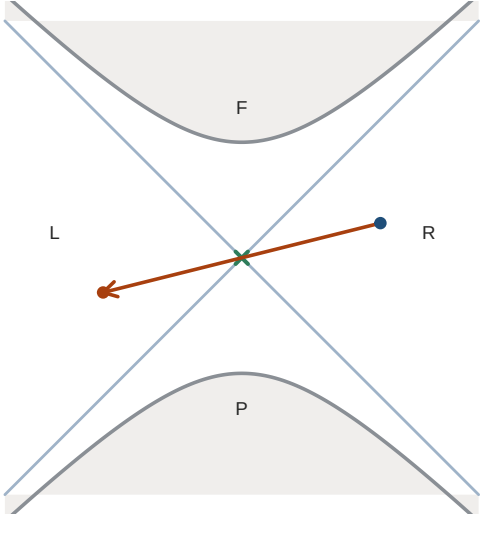
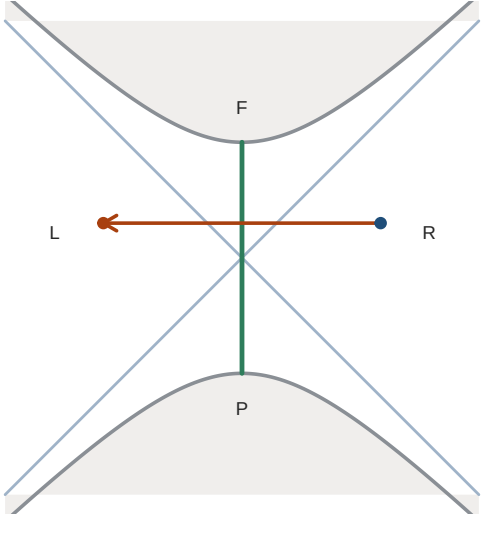
three involutions of the eternal hole, and only one can be the fold

N: $(U,V) \to (-U, -V)$



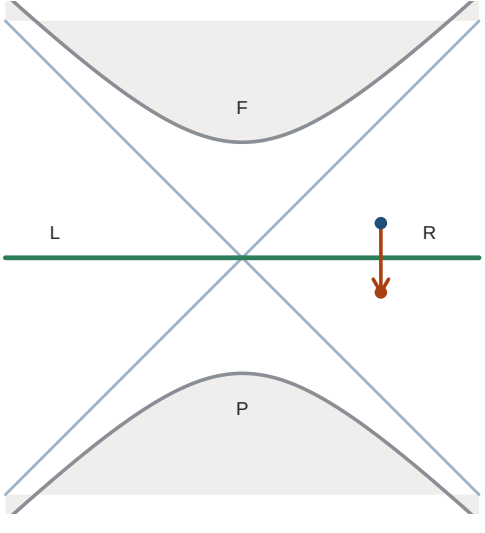
reverses time orientation and carries our exterior to the other one: the fold

S: $(U,V) \to (V, U)$



preserves time orientation, so CPT excludes it: this is the RP3 geon

SN: $(U,V) \to (-V, -U)$



sends each exterior to itself, so there is no mirror sheet for it to make